

JAYESH KOLI

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🌐 [LinkedIn](#) 🐙 [GitHub](#) 📁 [Portfolio](#)

OBJECTIVE

AI/ML Developer focused on building real-world machine learning projects, with practical experience in data preprocessing, model evaluation, and handling production-level data challenges like class imbalance.

EDUCATION

B.Tech (Computer Science and Engineering)

SVKM's College of Engineering, Daivad

Shirpur

[2024 – 2028] | Pursuing

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript.

AI/ML: NumPy, Pandas, Matplotlib, Scikit-learn.

Machine Learning: KNN, Linear Regression, Train-Test Split, Random Forest, Logistic Regression, SVM, Naive Bayes, NLP (Basics), Feature Scaling, Data Preprocessing, Model Evaluation.

Database: MySQL.

Deployment: Flask, FastAPI (REST API), Docker.

Version Control: Git, GitHub.

Development Tools: VS Code, Replit, Jupyter Notebook, Anaconda.

SOFT SKILLS

- Demonstrated **leadership** in guiding team projects and resolving challenges.
- **Adaptable** and quick to learn new tools, technologies, and environments.

PROJECTS

- **HUMAN ACTIVITY RECOGNITION API**
 - Trained and compared SVM classifiers (Linear vs RBF kernel) to classify human activity from smartphone sensor data (561 features, 6 classes), achieving 99.02% accuracy.
 - built a FastAPI service with Pydantic validation, containerized with Docker, and deployed live on Render.
 - **Technologies:** Python, Scikit-learn, FastAPI, Pydantic, Docker, Pandas.
 - **GitHub:** [Model training](#) | [API deployment](#)
 - **Live API:** <https://har-fastapi.onrender.com/docs>
- **CUSTOMER CHURN PREDICTION**
 - Built an end-to-end churn prediction model using Logistic Regression on real-world telecom data with inconsistent categories, missing values, and class imbalance (~27% churn).
 - Designed a preprocessing pipeline using ColumnTransformer and Pipeline to prevent data leakage between train and test sets.
 - Diagnosed misleading accuracy caused by class imbalance; applied `class_weight="balanced"`, improving churn recall from 0.59 to 0.83.
 - Evaluated using confusion matrix, precision, recall, and F1-score instead of relying on accuracy alone.
 - **Technologies:** Python, Pandas, Scikit-learn (Pipeline, ColumnTransformer, LogisticRegression).
 - **GitHub:** <https://github.com/jayeshkoli-1402/Customer-Churn-Prediction>
- **NEWS ARTICLE CLASSIFIER (NLP)**
 - Built a multiclass text classifier on the News Category Dataset using TF-IDF vectorization, achieving 71.01% accuracy with Logistic Regression (vs. Linear SVM, Naive Bayes).
 - Designed preprocessing and evaluation pipeline; exported trained model for reuse in deployment.
 - Deployed as a live REST API using FastAPI with interactive documentation.
 - **Technologies:** Python, Scikit-learn, Pandas, TF-IDF, FastAPI.
 - **GitHub:** [Model training](#) | [API deployment](#)
 - **Live API:** <https://nac-api-opmx.onrender.com/docs>

CERTIFICATIONS

- The Joy of Computing using Python — NPTEL (IIT Madras), Score: 90%.